

Algebraic Topology: a beginner's course

This lecture:

**AlgTop28: Covering spaces
and fundamental groups**

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Lecture 28 Covering spaces + fundamental groups



C_2'
 $C_1 \sim C_2'$

$C_2 \sim C_1'$

C_1'

α

Lifting Homotopy Lemma

If $p: X \rightarrow B$ is a covering space and α, β are homotopic paths in B from a to b a homotopy H , then there is a unique homotopy $\tilde{H}$ from the lifts $\tilde{\alpha}, \tilde{\beta}$ of α, β to X starting at some x with $p(x) = a$.

Proof. We lift α_s for each $s \in (0, 1)$ to $\tilde{\alpha}_s$ i

fundamental groups

B covering space.

$x \in X$ with $p(x) = a$

induced homomorphism

$$\pi(X, x) \rightarrow \pi(B, a)$$

$$p_*([\alpha]) = [p \circ \alpha]$$

Thm. If $p: X \rightarrow B$ is a covering space,
the $p_*: \pi(X, x) \rightarrow \pi(B, b)$ is injective
 $\Leftrightarrow$ it is l.i. $\Leftrightarrow \ker p_* = e$

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next in the series:

**AlgTop29: Universal
covering spaces**

Thanks to Nguyen Le for filming

